



Leica DMI3000 B, DMI4000 B, DMI6000 B

Instructions

Living up to Life

Leica
MICROSYSTEMS

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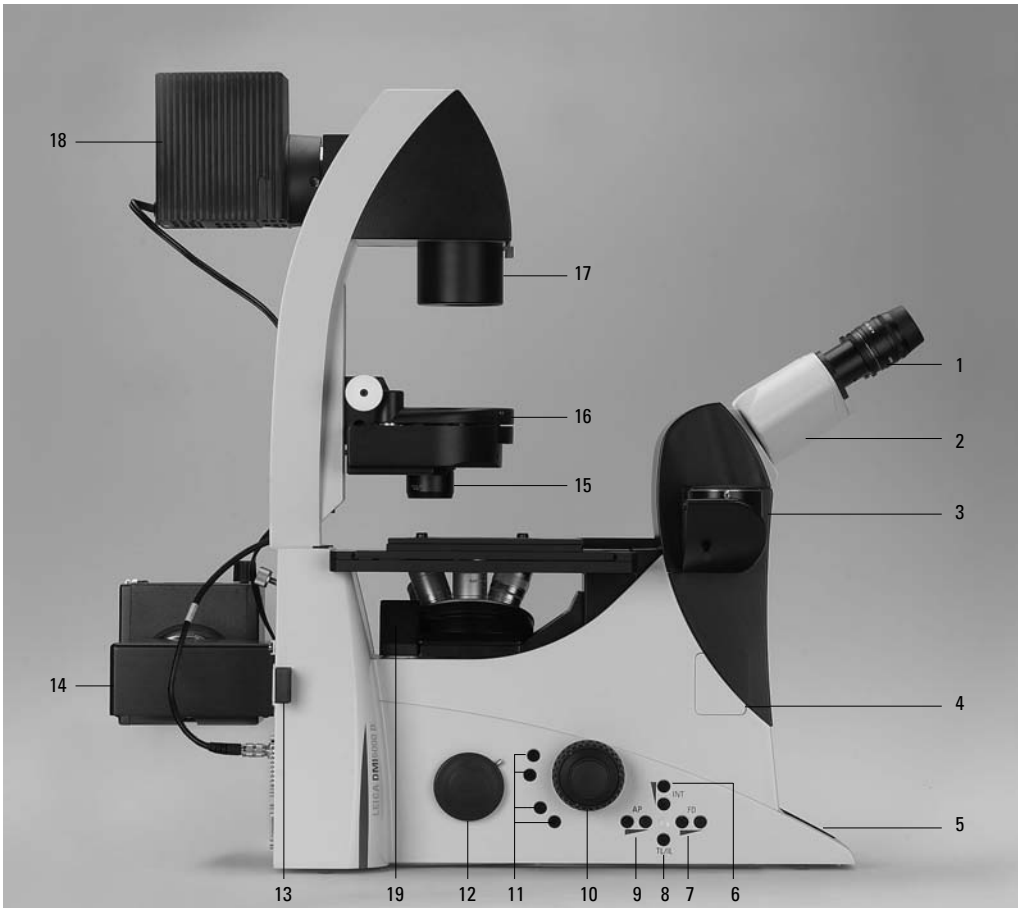


Fig. 1 Left side, Leica DMI4000 B and DMI6000 B

- | | |
|---|---|
| 1 Eyepiece | 11 Variable function buttons |
| 2 Eyepiece tube | 12 Left side port |
| 3 Top port | 13 Booster lens |
| 4 Intermediate pupil interface | (Leica DMI6000 B fluorescence microscopes only) |
| 5 LeicaScreen | 14 Lamp mount (fluorescence microscopes only) |
| 6 Light intensity | 15 Condenser head |
| 7 Field diaphragm | 16 Condenser base |
| 8 TL/IL switching | 17 Field diaphragm |
| 9 Aperture diaphragm | 18 Transmitted light lamp housing |
| 10 Focus wheel (motorized Leica DMI6000 B,
manual (fine and coarse) Leica DMI4000 B) | 19 DIC objective prism disk |

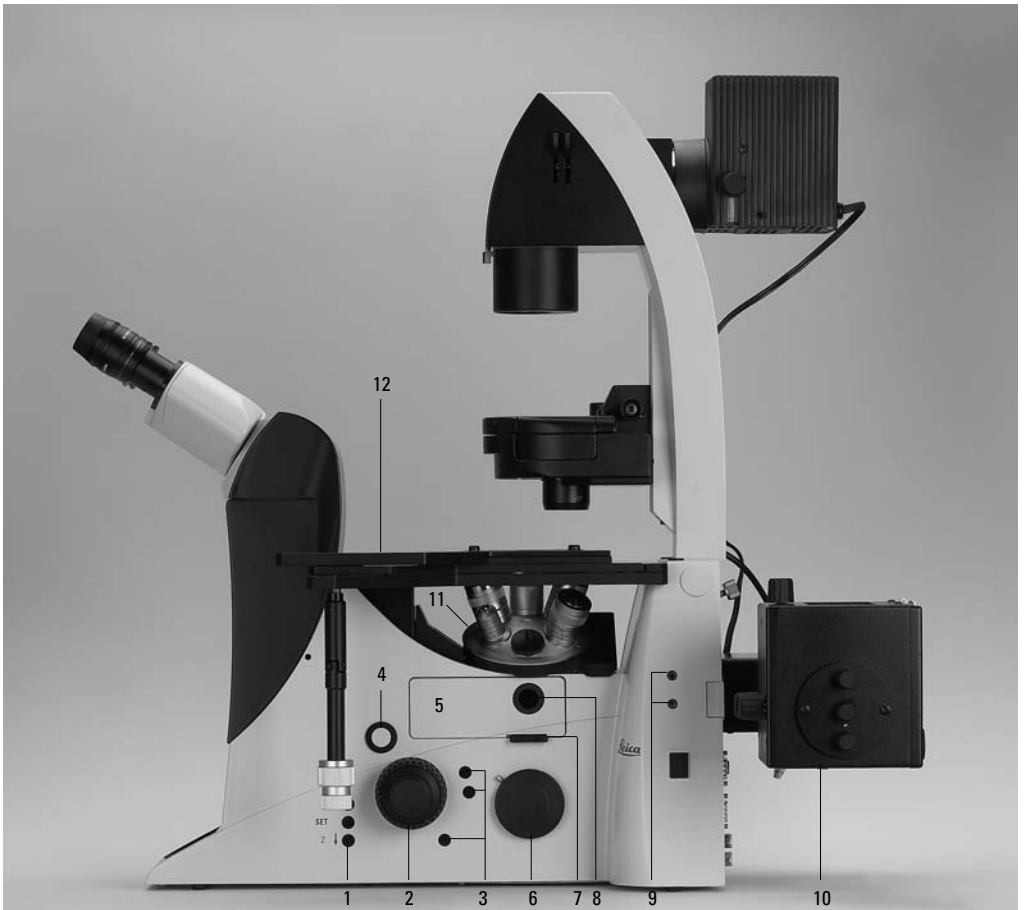


Fig. 2 Right side Leica DMI4000 B and DMI6000 B

- | | |
|---|--|
| <p>1 E-Focus buttons (Leica DMI6000 B only)</p> <p>2 Focus wheel (motorized Leica DMI6000 B, manual (fine) Leica DMI4000 B)</p> <p>3 Variable function buttons</p> <p>4 Opener for drawer (fluorescence microscopes only)</p> <p>5 Drawer (fluorescence microscopes only)</p> <p>6 Right side port</p> <p>7 Analyzer slot</p> | <p>8 Centering window (fluorescence microscopes only)</p> <p>9 Field diaphragm centering (fluorescence microscopes only)</p> <p>10 Incident light lamp housing (fluorescence microscopes only)</p> <p>11 Objective turret</p> <p>12 Stage with attachable mechanical stage</p> |
|---|--|

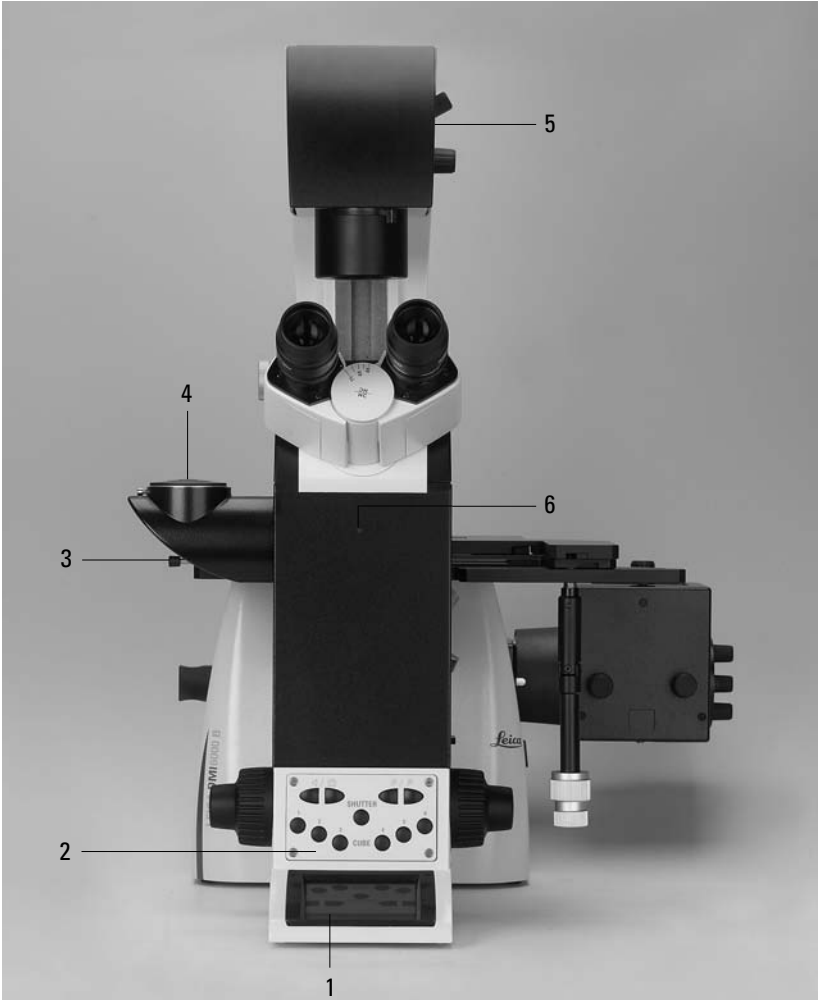


Fig. 3 Front view Leica DMI4000 B and Leica DMI6000 B

- 1 LeicaScreen
- 2 Front control panel
- 3 Port switching
- 4 Top port
- 5 Manual transmitted light filters
- 6 Bertrand lens centering

Fig. 3a Front control panel

- 1 Fluorescence cube
- 2 Shutter
- 3 100% light to all eyepieces
- 4 Port selection
- 5 Magnification selection
- 6 1x tube lens

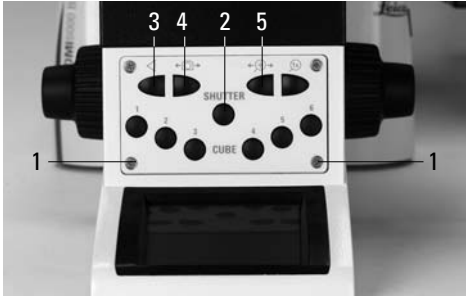


Fig. 3b SmartMove remote control module

- 1 Travel in x
- 2 Travel in y
- 3 Focus
- 4 Variable function buttons (pre assigned at factory)

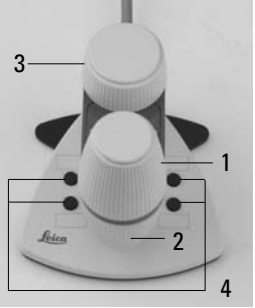


Fig. 4 General view Leica DMI4000 B and Leica DMI6000 B with SmartMove remote control module



7. Start-up

7.1 Functional Principle (Leica DMI4000 B and Leica DMI6000 B)

Thanks to its intelligent automation, the Leica DMI4000 B and DMI6000 B can be controlled using a variety of control elements.

1. Intelligent Automation

- Switching between contrast methods at the touch of a button. Light rings, DIC prisms, etc. are automatically positioned in the beam path.
- The microscope recognizes the selected objective and associated contrast method. The intensity (INT), aperture diaphragm (AP) and field diaphragm (FD) are always set to suitable values.
- The INT, AP and FD values are always based on the currently activated illumination axis (transmitted light or incident light).
- The INT, AP, and FD values can be adjusted individually. Manual adjustments overwrite the previous settings. The current setting is stored and is retained from one session to the next when power is switched off.

2. Controls

- SmartMove knobs
for stage and focus control
- Fixed function buttons on stand
for INT, AP, and FD, as well as for switching between transmitted light and incident light axis
- Variable function buttons on stand, SmartMove, STP6000
These function buttons have functions suitable to the configuration of your microscope assigned to them at the factory. The functions can be reprogrammed and/or adapted to your specific requirements, however.
- Complete control of microscope and camera via software
(Leica Application Suite (LAS/LAS AF))

7. Start-up

Function (DMI4000 B and DMI6000 B)	Fixed Function buttons Stand		Variable Function buttons Stand		SmartMove Function buttons		Rotary knobs	Software and STP6000
	4000	6000	4000	6000	4000	6000	4000 6000	4000/6000
Select contrast method	-	-	+	+	+	+	- -	+
Change transmitted light/incident light axis	+	+	-	-	-	-	- -	+
Change to objective	-	-	-	+	-	+	- -	+
Teach-in parfocality	-	-	-	-	-	-	- -	+
Change operating mode (dry/imm)	-	-	+	+	+	+	- -	+
Illumination Manager	+	+	+	+	+	+	- -	+
Magnification changer (motorized)	+	+	-	-	-	-	- -	+
Focusing	-	+	-	-	-	-	- ⁴¹⁾	+
Set stops	-	+	-	-	-	-	- -	+
Go to stop	-	+	-	-	-	-	- -	+
Change step increment (coarse/fine)	-	-	-	+	-	+	- -	+
XY stage positioning	-	-	-				+	+
Change speed	-	-	-	-	-	-	- -	+
Stage positions (store/go to)	-	-	-	-	-	-	- -	+
Change to filter/reflector cube	+		+	(+)	+	+	- -	+
Side and bottom port (DMI6000 B only)	+		(+)		+		- +	
DIC fine adjustment	+	+	-	-	-	-	- -	+
Adapted Focus Control (AFC)	-	+	-	+	-	+	- -	+

+ always possible

(+) optional

- not possible

⁴¹⁾ Focusing alternatively via wheels

Possible Assignments for Variable Function Buttons on Stand and SmartMove For Leica DMI4000 B and Leica DMI6000 B:

Function button	Function
BF	Bright field transmitted light
PH	Phase contrast transmitted light
ICT	Interference contrast, transmitted light
DF	Dark field transmitted light
IMC	Integrated modulation contrast
POL	Polarization transmitted light
CHANGE TL 	Cycle through all contrast methods
INT ↑	Increase intensity (transmitted light)
INT ↓	Reduce intensity (transmitted light)
AP ↑	Open aperture diaphragm (transmitted light)
AP ↓	Close aperture diaphragm (transmitted light)
FD ↑	Open field diaphragm (transmitted light)
FD ↓	Close field diaphragm (transmitted light)
SHUTTER TL	Open/close TL shutter
TL FLT 1	Enable/disable transmitted light filter at position 1
TL FLT 2	Enable/disable transmitted light filter at position 2
FLUO	Fluorescence (last filter cube)
CUBE 1-6	Select filter cube in position 1-6
CHANGE CUBE CW	Change cube clockwise (1 → 4)
CHANGE CUBE CCW	Change cube counterclockwise (4 → 1)
INT FLUO ↑	Increase intensity (fluorescence)
INT FLUO ↓	Reduce intensity (fluorescence)
FD FLUO ↑	Open field diaphragm (fluorescence)
FD FLUO ↓	Close field diaphragm (fluorescence)
CHG FW	Toggle filter functions
IFW	Activate external filter wheel
ExMan	Activate Excitation Manager
SHUTTER FL	Open/close fluoshutter
COMBI 	Combination method (PH fluorescence or ICT fluorescence)
CHANGE COMBI 	Cycle through all combination methods
CHANGE OBJ CW	Cycle through objectives clockwise
CHANGE OBJ CCW	Cycle through objectives counterclockwise
Z FINE	Activate fine focusing (Leica DMI6000 B only)
Z COARSE	Activate coarse focusing (Leica DMI6000 B only)
XY PRECISE	Activate precise stage
XY FAST	Activate fast stage
BTP ON/OFF	Bottom port on/off (Leica DMI6000 B only)
DRY/IMM	Switch dry/immersion
CHANGE FLT	Switch TL filter
CHANGE CS	Switch to confocal application
OBJ 1-6	Select objective at position 1-6
MEM 1-6	Memory activated stored functions
AFC ON/OFF	Turns AFC on or off
AFC HOLD	Holds current position

7.2 Switching on the Microscope

Leica DMI3000 B:

- Switch on the microscope's power at the On/Off switch. The signal lamp is lit when the instrument is ready. (For the Leica DMI3000 B please continue at 7.4. Function Buttons on the Stand)

Leica DMI4000 B and Leica DMI6000 B:

- Switch on the power of the electronics box at the On/Off switch (80.1). The signal lamp (80.2) is lit green when the unit is ready. All motorized microscope components will then run through an initialization phase.



Note:

If a PC is connected, switch on **the electronics box first**, and **then the computer**.

After the initialization (Fig. 81) is complete, the LeicaScreen will display the microscope's current settings (Fig. 82).

Fig. 80

front side

Leica CTR6000

- 1 On/Off switch
- 2 Signal lamp



If a component has not been installed correctly, the LeicaScreen will display an error message. See Troubleshooting chapter, → p. 105.

Components such as diaphragms, condensers, light, and phase rings have been pre-centered at the factory. It may be necessary to correct the centering after the microscope has been transported and assembled.

Before performing the required steps, please familiarize yourself with the LeicaScreen and the controls.



Caution!

After turning on the gas discharge lamp, the burner must be immediately adjusted. Therefore, **do not** turn on the power supply unit yet. First, work in transmitted light in order to familiarize yourself with the microscope's controls.

Fig. 81

LeicaScreen
Initialization



Fig. 82

LeicaScreen
after
Initialization

	FLUO>DIC	
	40x Obj. IMM	
	1.5x MagCh.	Σ 600x
	INT 100%	B G ↺+1 ↻+2
	AP 33	FD 30
	80%	20%
	- 0.55 mm	coarse

7.3 The LeicaDisplay (Leica DMI 4000 B and DMI 6000 B)

The screen displays the microscope's current settings. The content of the display depends on the features of the individual microscope.

For information on the abbreviations used, please turn to the table of abbreviations → p. 113.

The screen has a number of areas and lines.

- | | |
|---------|-------------------------------|
| Line 1: | contrast method |
| Line 2: | objective/magnification |
| Line 3: | illumination/diaphragms |
| Line 4: | active ports |
| Line 5: | focus/stops (DMI 6000 B only) |

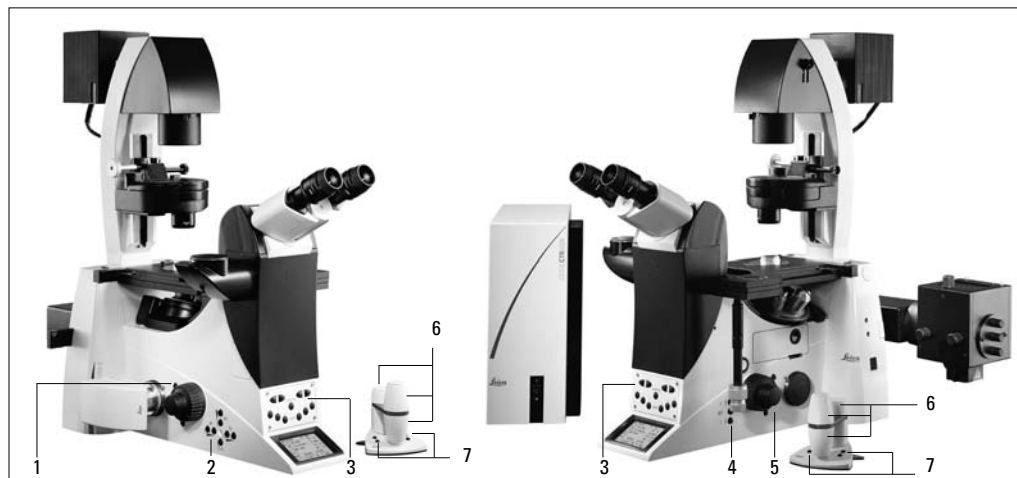
The content of the display changes according to the active function.

Pictograms

	Contrast method
	Objective/ Magnification
	Illumination Diaphragm
	Ports/Eyepiece
	Focus/stops (DMI6000 B only)

Fig. 83 Arrangement of the function buttons – overview

- 1 Four variable function buttons
- 2 Illumination Manager
- 3 Front control panel
- 4 Focus buttons (DMI6000 B only)
- 5 Three variable function buttons
- 6 SmartMove knobs
- 7 SmartMove function buttons



7. Start-up

The motorized DMI stands can be controlled using the function buttons on the stand, the remote control SmartMove or the STP6000.

7.4 The Function Buttons on the Stand

Leica DMI3000 B:

- Focus wheels: the left-hand focus wheels can be used for both coarse and fine focusing; the right-hand focus wheel for fine focusing only (a version of the Leica DMI3000 B with mirrored focus controls is also available)
- Light intensity: the transmitted light intensity can be adjusted continuously from 0 to 12 V using the potentiometer at the lower left of the front of the microscope stand.

For the Leica DMI3000 B please continue at 7.6. Illumination.

Leica DMI4000 B and Leica DMI6000 B:

A number of function buttons are located on both sides of the stand. These can be broken down into fixed and variable buttons. The variable function buttons have different functions depending on the features of the individual microscope.

Fixed function Buttons on the Left Side

The **TL/IL** button (84.1) toggles between the incident-light and transmitted light axis. The contrast method last used with a given axis is restored when switching.

The **INT** buttons (84.3) adjust the light intensity. The adjustment can be made in coarse or fine steps. Pressing both **INT** buttons at the same time toggles between coarse and fine adjustment.

The **AP** buttons (84.2) for the aperture diaphragm and **FD** (84.4) for the field diaphragm open and close their respective diaphragms.

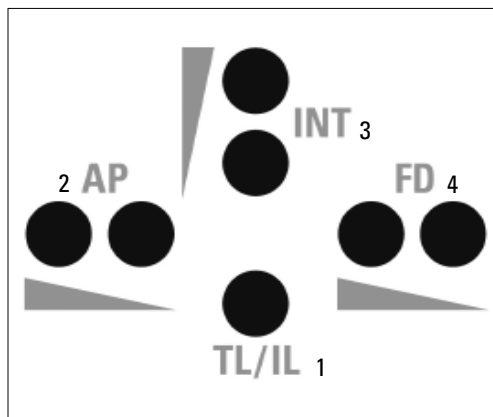


Note:

Changes to the light intensity as well as aperture and field diaphragm settings are stored for the individual objectives and contrast methods.

Fig. 84 Fixed function buttons (left side of stand)

- 1 Toggle transmitted light/incident light
- 2 Aperture diaphragm
- 3 light intensity
- 4 field diaphragm



Variable Function Buttons on the Stand

The variable function buttons are assigned functions at the factory that are appropriate to the features of your microscope. They are labeled accordingly. For details on button assignments, please refer to the included identification sheet. For information on the abbreviations used, please refer to the list → p. 63.



Note:

The Leica Application Suite (LAS) software is required for changing the button assignments.

Possible functions*:

BF	CHANGE CUBE CW
PH	CHANGE CUBE CCW
ICT	INT FLUO ↑
DF	INT FLUO ↓
IMC	FD FLUO ↑
POL	FD FLUO ↓
CHANGE TL ●	CHG FW
INT ↑	IFW
INT ↓	ExMan
AP ↑	COMBI ●
AP ↓	CHANGE COMBI ●
FD ↑	CHANGE OBJ CW (only DMI6000 B)
FD ↓	CHANGE OBJ CCW (only DMI6000 B)
SHUTTER TL	Z FINE (only DMI6000 B)
TL FLT 1	Z COARSE (only DMI6000 B)
TL FLT 2	XY PRECISE
FLUO	XY FAST
CUBE 1	BTP ON/OFF (only DMI6000 B)
CUBE 2	DRY/IMM
CUBE 3	CHANGE FLT
CUBE 4	CHANGE CS
CUBE 5	OBJ 1-6
CUBE 6	MEM 1-6
AFC ON/OFF	Turns AFC on or off
AFC HOLD	Holds current position

* See page 63 for abbreviations

Fig. 85 Function buttons (left side of stand)

- 1 Variable function buttons
- 2 Open/close aperture diaphragm
- 3 TL/IL switching
- 4 Open/close field diaphragm
- 5 Increase/decrease light intensity

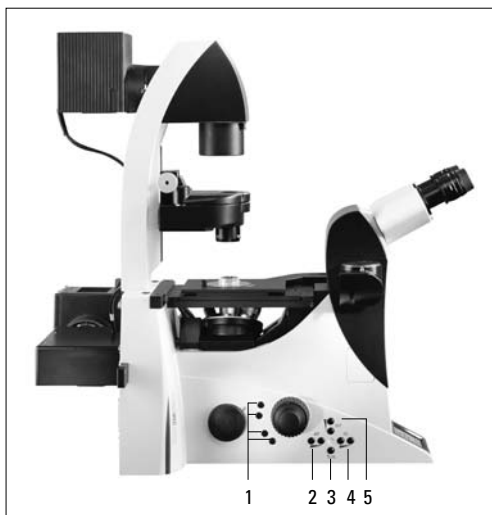
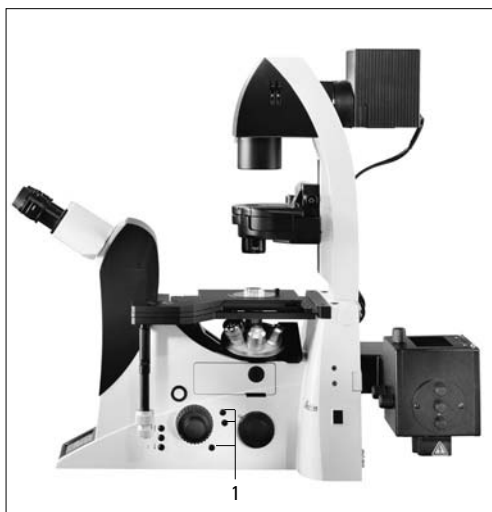


Fig. 86 Function buttons (right side of stand)

- 1 Variable function buttons



Function Buttons on the Front Panel (Fig. 87)

 100% of the light goes to the eyepiece (87.1).

←  → Toggle function for the side ports (87.2). This function depends on the individual microscope configuration.

Note:
Switching to the bottom port:
via the variable function buttons (Leica DMI6000 B only), switching to top port: manually.

SHUTTER Opens and closes the shutter (87.3).

←  → Switches between the possible magnifications of the magnification changer (87.4).

 The magnification changer is set to the magnification 1x (87.5).

CUBE The CUBE 1 to CUBE 6 (87.6) buttons permit the direct selection of individual filter cubes, provided the selected cube is valid for the selected method.
Press the CUBE 3 and CUBE 4 buttons at the same time to display the assignments of the variable function buttons. To reset the display, press the buttons again or wait 3 seconds.

Focus buttons (Fig. 88) (DMI6000 B only)

Z↑ Moves the Z drive in the indicated direction.
Z↓

SET + Z↑ Sets the upper focus stop.

SET + Z↓ Sets the lower stop.

Fig. 87 Front control panel

- 1 100% light to eyepiece
- 2 Toggle ports
- 3 Shutter
- 4 Switch between subsequent magnifications
- 5 Subsequent magnification 1x
- 6 Selecting filter cubes

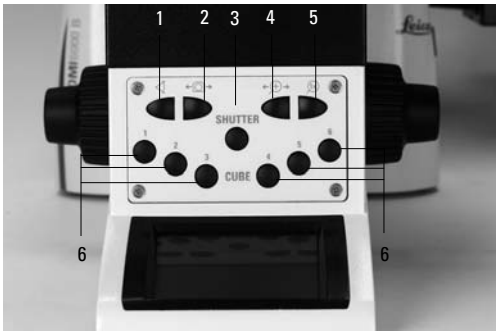
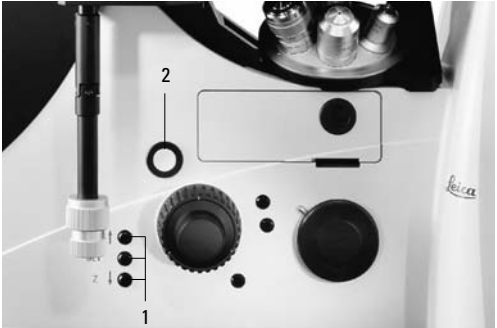


Fig. 88

- 1 Focus control buttons
- 2 Open filter drawer



Shutter button + Cube buttons 1-6 (Leica DMI4000 B and Leica DMI6000 B only)

The selected cube is moved to the loading position for replacement. „Load“ appears on the screen. After pressing the button (88.2) the drawer is opened and the cube can be changed. The next filter cube will be moved to the loading position after the drawer is closed.

7.5 The SmartMove Remote Control Module

SmartMove knobs

(Leica DMI4000 B and Leica DMI6000 B)

Use the knobs 89.1 and 89.2 to move the stage in X and Y directions.

The image is focused using the knob 89.3 (Leica DMI6000 B only).

The height of the knobs can be adjusted to a comfortable working position by turning 89.4.

Variable function buttons on SmartMove

The variable function buttons are assigned functions at the factory that are appropriate to the features of your microscope. They are labeled accordingly. For details on button assignments, please refer to the included identification sheet.

For information on the abbreviations used, please refer to the list → p. 63.



Note:

The Leica Application Suite (LAS) software is required for changing the button assignments.

7.6 Illumination

7.6.1 Transmitted light

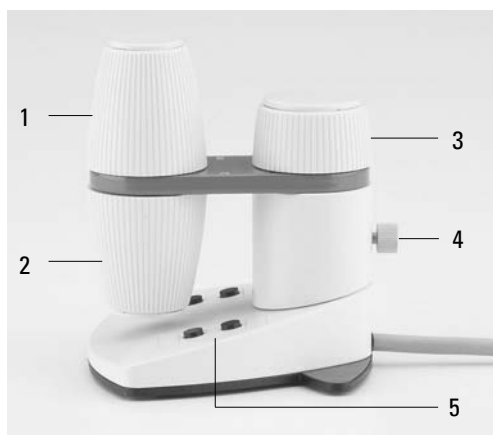
If your microscope has not yet been set up for Koehler illumination, please continue with the „Koehler Illumination“ section.

Leica DMI3000 B:

- If necessary, adjust the TL bright field position at the filter slider.
- Select an objective with moderate magnification (10x–20x).
- Set the condenser to the bright field position.
- Place a specimen on the stage.
- Focus on the specimen using the focus wheels.
- Adjust the light intensity.
- Close the field diaphragm manually until the edge of the diaphragm appears in the field of view.

Fig. 89 SmartMove remote control module

- 1 travel in x
- 2 Travel in y
- 3 Focus
- 4 Individual adjustment of button height
- 5 Variable function buttons (factory preset)



8.2 Contrast Methods

All of the contrast methods of the Leica DMI4000 B and Leica DMI6000 B can be selected and controlled via the variable function buttons and the Leica Application Suite (LAS). The only exceptions are methods that involve components requiring manual control (e.g. systems with manual analyzers). The following section describes the use of the function buttons on the stand. For instructions on the use of the software, please refer to the separate manual.

Contrast methods for the Leica DMI3000 B are controlled via the manual condenser, the manual objective turret, as well as turning knobs and sliders at the microscope.

8.2.1 Bright Field (TL)

Leica DMI3000 B:

- If necessary, adjust the TL bright field position at the filter slider.
- Set the condenser to the bright field position.
- Remove all other optical components such as analyzers, polarizers or IC prisms from the beam path.
- Insert a transmitted light specimen.
- Select your objective
- Set the brightness at the light potentiometer
- Focus the image with the focus wheels.

Leica DMI4000 B and Leica DMI6000B:



Note:

If all positions of the filter turret are occupied, filter cube „A“ can be swapped for filter cube „A-TL“ using the Leica Application Suite (LAS/LAS AF). TL contrast methods are possible with that filter cube.

8. Operation

- Use the **TL/IL** function button to switch to transmitted light (TL).
 - Select the **BF** (bright field) contrast method by pressing the variable button **BF**.
Alternatively: press the variable button **CHANGE TL** .
- (For details on button assignments, please see the identification sheet.)
- BF** will appear on the LeicaScreen.
- Motorized condensers will now move to the bright field position. Coded condensers must be switched manually.
- The fluorescence filter turret will automatically go to an empty position or to the „A-TL“ filter cube.
- Insert a transmitted light specimen.
 - Rotate an appropriate objective into place.
 - Focus the image with the knob on the Smart-Move or the focusing wheel and adjust the intensity with the **INT** function buttons.

Fig. 108 Function buttons (left side of stand)

- 1 variable function buttons
- 2 Open/close aperture diaphragm
- 3 TL/IL switching
- 4 Open/close field diaphragm
- 5 Increase/decrease light intensity

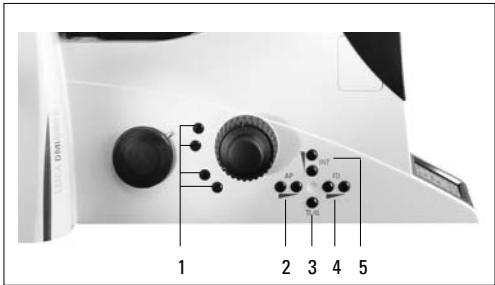
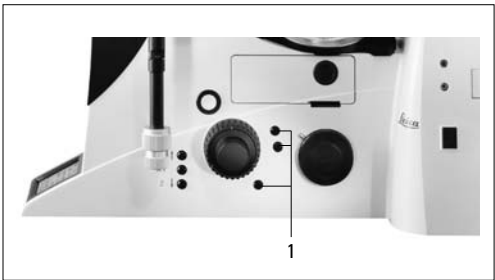


Fig. 109 Function buttons (right side of stand)

- 1 variable function buttons



8.2.2 Phase Contrast (TL) (integrated phase contrast, see 8.2.6)

Leica DMI3000 B:

- If necessary, adjust the TL bright field position at the filter slider.
- Select a phase contrast objective.
- Select the suitable light ring on the condenser.
- Open the aperture of the condenser completely.
- Remove all other optical components such as analyzers, polarizers or IC prisms from the beam path.
- Insert a phase contrast specimen.
- Set the brightness at the light potentiometer
- Focus the image with the focus wheels.

Leica DMI4000 B and Leica DMI6000 B:

- Use the **TL/IL** function button to switch to transmitted light (TL).
- Select the **PH** (phase contrast) contrast method by pressing the variable button **PH**.
Alternatively: press the variable button **CHANGE TL** .
(For details on button assignments, please see the identification sheet.)
PH will appear on the LeicaScreen.
Motorized condensers will now switch to the correct light ring. Coded condensers must be switched manually.
- Insert a transmitted light specimen.
- Rotate an appropriate objective into place.
Objectives that are suitable for phase contrast are engraved with PH.
- Focus the image with the knob on the Smart-Move or the focusing wheel and adjust the intensity with the **INT** function buttons.



Note:

When selecting the phase contrast method, the aperture diaphragm is opened fully and can not be adjusted.

8.3 Fluorescence

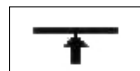
Leica DMI3000 B:

The filter slider (5a.9, S.24) is used to operate the fluorescence module.

- Pull the filter slider out completely to open the beam path.
- Push the filter slider into the middle position (1st detent) to bring the blue filter into the beam path.
- Insert the filter slider fully in order to block the beam path (shutter position).
- The fluorescence illumination is controlled by the rotary knob (5a.10, S.24).
- The filter cubes are swiveled manually into the beam path by turning the incident light turret disk.

Leica DMI4000 B and Leica DMI6000 B:

- Use the **TL/IL** function button to switch to fluorescence **FLUO**.
- Place a specimen on the stage and select a suitable objective.
- The current fluorescence filter cube will be displayed on the LeicaScreen.
- You may protect your specimen from fading by closing the incident light shutter. To do so, press the **SHUTTER** button (87.3) on the front panel. The following pictogram will appear on the LeicaScreen:



► Changing the fluorescence filter cube

- Fixed function buttons on the front panel: **CUBE 1** to **CUBE 6** or **Cube CCW**
- Variable function buttons on the front panel and SmartMove: **CUBE CW** or **CUBE CCW**
- Leica Application Suite (LAS) Software

- Focus the image with the knob on the SmartMove or the focusing wheel and adjust the intensity with the **INT** function buttons.

- For multiple fluorescence, we recommend using the Excitation Manager and/or the ultrafast internal filter wheel. Excitation wavelengths can thus be changed in milliseconds. They are controlled by the function buttons.

[illegible]

8.4 Combination Methods

(Leica DMI4000 B and DMI6000 B)

Up to two combination methods are possible depending on the features of the individual microscope:

FLUO/PH and FLUO/DIC

- Select the combination method by pressing the variable button **COMBI** . Alternatively: press the variable button **CHANGE COMBI** . (For details on button assignments, please see the identification sheet.) The content of the display changes accordingly.
- Place a specimen on the stage and select a suitable objective.
- Select the desired filter cube using the fixed function buttons on the front panel.
- The illumination settings for the fluorescence and transmitted light axes can be adjusted separately.
- Toggle the illumination axes with the **TL/IL** function button. The content of the LeicaS-screen changes accordingly.

FLUO > DIC

The transmitted illumination is activated.

FLUO < DIC

The fluorescence illumination is activated.



Note:

The manual analyzer (Fig. 110) must be used for the FLUO/DIC method as described in Chapter 8.2.5, p. 88.

8. Operation

8.5 Focusing

Leica DMI3000 B and Leica DMI4000 B:

The left-hand focus wheels can be used for both coarse and fine focusing; the right-hand focus wheel for fine focusing only (a version of the Leica DMI3000 B with mirrored focus controls is also available)

Leica DMI6000 B:



Note:

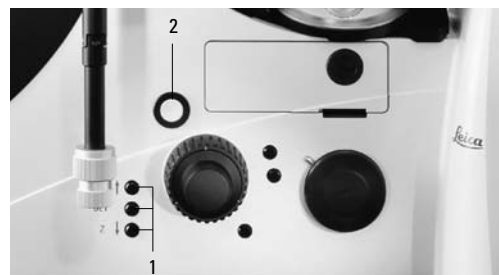
The parfocality teach-in has already been performed at the factory. However, it may be necessary to perform another teach-in after installing the objectives when setting the microscope up. We recommend checking parfocality before setting the stops and performing a teach-in with the Leica Application Suite (LAS) if necessary.

Focusing the image

The focusing is controlled by the knobs (116.3, p. 101) on the SmartMove remote control module.

Abb. 114

- 1 Focus operating keys
- 2 Open filter drawer



Alternatively, use the focus wheels on either side of the stand.

The current Z position is shown on the LeicaScreen. In the case of motorized stages, the Z drive will travel to its lowest position prior to the stage initialization when switching the microscope on.

The focus buttons **Z↑** and **Z↓** on the right side of the stand (Fig. 114) permit fast focusing or lowering of the objectives.

Setting stops

Set the lower focus stop by pressing and holding the **SET** button and pressing the **Z↓** button as well.

The display will show ▼.

Pressing the button combination again will delete the stop.

The display will show ▼.

The lower focus stop can also be set using the Leica Application Suite (LAS).

The **lower stop** is the same for all objectives and can not be traversed.

In addition, a **focus position** that can not be traversed can also be set.

To do so, press and hold the **SET** button and press the **Z** button as well.

The display will show ✕.

Pressing the button combination again will delete the stop.

The display will show ✕.

The focus position can also be set using the Leica Application Suite (LAS).

Set the focus position for the dry objective at the highest magnification. The focus positions will then be set automatically for all other objectives, taking parfocality and working distances into account.

► Set the stops via

- fixed function buttons on stand
- Leica Application Suite (LAS) Software

Summary of pictograms

- ▼ lower focus stop not set
- ▼ lower focus stop set
- ✂ focus position not set
- ✂ focus position set

Going to the stops

Go to the lower stop by pressing and holding the **Z↓** button.

Go to the focus position by pressing and holding the **Z↑** button.

These functions can be assigned to variable function buttons on the stand or SmartMove, or they can be controlled via software.

► Go to stops via

- fixed function buttons on stand
- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS) Software



Note:

When going to the stops with the **Z↑** and **Z↓** buttons, hold the button until the stop has been reached.

Setting the step increments

It is possible to toggle between **Fine** and **Coarse** step increments.

The **Fine** value varies to suit the current objective. Suitable values have been predefined. The assignments can be changed with the Leica Application Suite (LAS). When selecting **Coarse**, the positioning speed is the same for all objectives. **Coarse** corresponds to the maximum speed.



Note:

The assignment of a specific step increment to an objective not only applies to the Z drive, but also to the step increments assigned to the stage when **Precise** (→ p. 101) is selected.

► Switch between Fine and Coarse via

- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS) Software

Only for DMI6000 B with AFC (Adaptive Focus Control)

AFC actively holds a pre-defined focus position over time. This feature is especially useful, if e.g. during a time-lapse experiment at 37°C the climate chamber has to be opened and a drop of the temperature may occur.

Activate the AFC function, focus on your specimen either with the hand wheel on the microscope stand or at the SmartMove and store the current focus position as hold position.

AFC can be controlled either by.

- Variable function keys at the stand or SmartMove
- STP6000
- Software (Leica LASAF)

8.6 Tubes



Note:

Close any unused tube openings, as otherwise stray light can interfere with observation.

Adjusting the viewing distance

- Adjust the viewing distance of the eyepieces so that a congruent total image is seen (Fig. 115).

Adjusting the viewing angle

- Ergotubes feature a tilting binocular section for a 30–45° viewing angle adjustment range.

Beam splitting in photo tubes

The beam splitting is set manually by pulling out a control bar.

	Observation	photo
☐	100%	0%
☐	0%	100%
alternatively	50%	50%
BL ☐	activation of Bertrand lens*	

8.7 Port selection

Leica DMI 3000 B and Leica DMI4000 B:
Manual shifter rod activates and deactivates the left-hand photo port.

	VIS	LEFT
☐	100%	0%
☐	20%	80%
alternatively:	0%	100%

Leica DMI 6000 B:

The  button on the front control panel switches 100% of the light to the eyepieces. Use the  button, also on the front control panel, to select the side ports.

► **Light distribution via**

► manual control bar



Fig. 115 Tube setting

Depending on the configuration, the screen will now display

- the active port (right or left) and
- the percentage of light going to the port (100%, 80%, 50%).

Optional Leica DMI 6000 B:

The bottom port selection function can be assigned to one of the variable function buttons on the stand or the SmartMove.

The top port can only be selected manually.

8.8 Eyepieces



Note:

The eyepiece's aperture protector must be removed or folded back, during microscopy while wearing eyeglasses. We recommend removing bifocals and spectacles with progressive-addition lenses when using the microscope.

- For the adjustable tubes with documentation output, choose the 100% VIS position.

Eyepieces with inlaid reticle

- Focus the reticle by adjusting the eyelens.
- Focus on the object through this eyepiece.
- Then, close that eye and focus on the object by adjusting only the second ocular.

Correction for Vision Problems

- With your right eye, look through the right eyepiece and bring the specimen into sharp focus.
- Then, with your left eye, view the same specimen and rotate the left eyepiece tube until the object is brought into sharp focus. Do not change the Z position in the process!

► Select ports via

- fixed function buttons on stand (side ports)
- variable function buttons on stand and SmartMove (bottom port)
- manual action (top port)



Note:

We recommend running a teach-in via the Leica Application Suite (LAS) software when using eyepieces not included in the scope of delivery. This will ensure that the total magnification shown in the LeicaScreen is correct.

8.9 Objectives

Changing objectives

Leica DMI3000 B and Leica DMI4000 B:

Select objectives manually with the objective turret.

The objective turret of the DMI4000 B is coded so that the selected objective is shown on the display.

Leica DMI6000 B:

The objectives can be selected with the function buttons on the stand or the SmartMove, or by manually turning the objective turret. When changing objectives manually, please ensure that the turret clicks into position.

The positions of the objectives in the objective turret have been specified at the factory and must be observed when installing the objectives. (→ also see Objectives, p. 43).

When selecting an objective, the microscope automatically selects:

- the optimal setting for the field diaphragm
- the optimal setting for the aperture diaphragm
- the light intensity for the current contrast method

The objective magnification and total magnification are displayed on the LeicaScreen.

- For **immersion objectives** use the appropriate immersion medium.

OIL: only use optical immersion oil according to DIN/ISO standards.
Cleaning → p. 109.

W: Water immersion.

IMM: Universal objective for water, glycerol, oil immersion.



Caution!

Follow safety instructions for immersion oil!

Color coding of objectives

The magnification of each objective is indicated by a color ring in accordance with DIN/ISO standards:

100x 125x 150x 160x	63x	40x 50x	25x 32x	16x 20x	10x	6.3x	4x 5x	2.5x	1.6x
white	dark-blue	light-blue	dark-green	light-green	yellow	orange	red	brown	gray

Immersion objectives are marked by an additional, lower color ring.

black oil or Imm (universal objective for oil, water or glycerin)

white water

orange glycerin

The various engraved markings of the objectives provide information on their applications:

black or bright field objectives,
dark blue strain-free
green phase contrast objectives,
strain-free

► Select objectives via

- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS) Software
- Manual selection

Changing the operating modes

„dry“ (DRY) and „immersion“ (IMM)

Each objective is assigned to a specific objective category:

- 1) Dry objectives (DRY)
- 2) Immersion objectives (IMM)



Note:

It is possible to use objectives for both operating modes.

The mode can be assigned in the Leica Application Suite (LAS).

Changing the operating mode

- First, select the operating mode (Imm or Dry) using the function buttons.
The operating mode may also be selected in the Leica Application Suite (LAS).
- The objective turret is lowered to its bottom stop. This is to permit the application of the immersion liquid when changing from a dry to an immersion objective. It also permits the removal of the liquid when changing to dry mode. The current objective remains in the beam path.

- Next, press the button for the objective you intend to use.



Note:

If the **Imm** or **Dry** operating mode buttons are pressed accidentally, the original mode can be restored by pressing the appropriate button.

► Change operating mode via

- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS) Software



Note:

When replacing objectives, you must perform a teach-in for the new objectives in the Leica Application Suite (LAS). A parfocality teach-in should also be performed.



Note:

For lockable immersion objectives lock these by pushing the front part upwards until it stops (approx. 2 mm). Then, after a gentle turning motion to the right, the objective is locked.

For objectives with corrective mounts turn the knurl to adjust the objective to the thickness of the cover glass.

8.10 Stages and Object Displacement

Leica DMI3000 B and Leica DMI4000 B:

The motorized stages are controlled via a separate control unit.

Leica DMI6000 B:

Object displacement using SmartMove

The positioning of the stage is controlled by the knobs (116.1, 116.2) on the SmartMove remote control module.

Setting the step increments

The positioning speed of the stage can be varied by switching between the **Fast** and **Precise** step increments.

When selecting **Fast**, the positioning speed is the same for all objectives.

The **Precise** speed varies to suit the current objective.

Storing and restoring stage positions

A variety of stage positions can be stored temporarily in the Leica Application Suite (LAS). The XY position is stored, not the Z position.

In addition to a loading position (Load), 5 stage positions can be set temporarily. When switching the microscope on, the stage will travel to a previously-defined starting position.

► **Temporarily store and restore stage positions via**

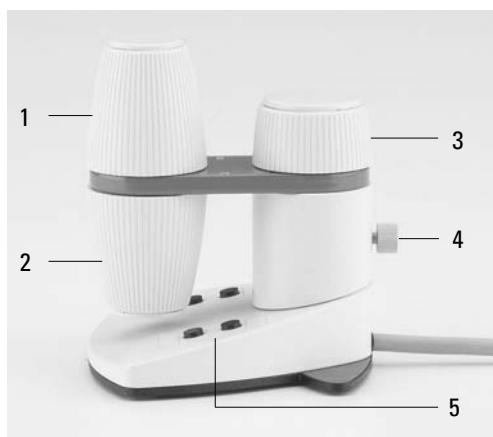
- Leica Application Suite (LAS) Software

► **Switch between Precise and Fast via**

- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS) Software

Fig. 116 SmartMove remote control module

- 1 travel in x
- 2 Travel in y
- 3 Focus
- 4 Individual adjustment of button height
- 5 Variable function buttons (factory preset)



8.11 Magnification Changer

Leica DMI3000 B:

A mechanical magnification changer can be used optionally.

Magnification factor: 1.6x.

A slider switches between 1x and the magnification factor. The mechanical magnification changer affects the eyepieces and the top port.

Leica DMI4000 B and Leica DMI6000 B:

A mechanical magnification changer can be used optionally. The following magnification factors are available: 1.5x, 1.6x and 2x

A slider switches between 1x and the magnification factor.

The mechanical magnification changer affects the eyepieces and the top port.

The selected factor is shown in the LeicaDisplay or the relevant window of the Leica Application Suite (LAS) and taken into account when calculating the total magnification.

Leica DMI4000 B and Leica DMI6000 B:

A motorized magnification changer can be used optionally. The following magnification factors may be selected: 1.5x, 1.6x, or 2x

The selected factor is displayed on the LeicaS-screen and in the relevant field of the Leica Application Suite (LAS), and is taken into account when calculating the total magnification. The motorized magnification changer affects all ports.

Pressing the left button (117.1) switches between the possible magnification factors; pressing the right button selects the factor 1x.



Note:

a microscope can not have both types (manual and motorized) of magnification changers.

Fig. 117 Front control panel

1 Function buttons for magnification changer



► **Change magnification via**

- fixed function buttons on stand
- Leica Application Suite (LAS) Software

8.12 Light sources

Leica DMI3000 B:

- Light intensity: the transmitted light intensity can be adjusted continuously from 0 to 12V using the potentiometer at the lower left of the front of the microscope stand.

- FLUO: The intensity can be adjusted in 5 fixed levels.

100% / 55% / 30% / 17% / 10%

(FIM=Fluorescence Intensity Manager)

If you continue turning to the left at 100% or to the right after 10%, a 0% position (shutter position) is set.

Leica DMI4000 B and Leica DMI6000 B:

- Adjust the intensity with the function buttons (118.4). The INT function buttons are always assigned to the currently active transmitted light (TL) or incident light (IL) axis.

- For TL and IL:

The setting can be made in coarse and fine steps. Pressing both INT (118.2) buttons at the same time toggles between coarse and fine adjustment. The light intensity displayed on the LeicaScreen changes accordingly.

Coarse adjustment: 0–20

Fine adjustment: 0–255

- The intensity is individually adjusted and stored for each objective and contrast method.

- FLUO: The intensity can be adjusted in 5 fixed levels.

100% / 55% / 30% / 17% / 10%

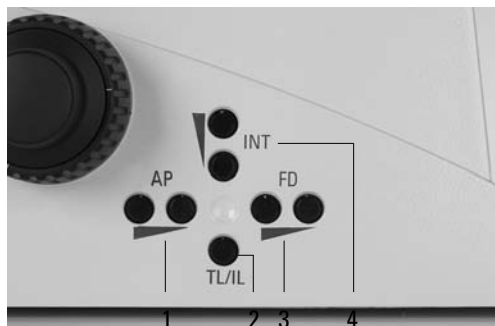
(FIM=Fluorescence Intensity Manager)

► Adjust intensity via

- fixed function buttons on stand
- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS) Software

Fig. 118 Fixed function buttons, left side of stand

- 1 Aperture diaphragm
- 2 Transmitted light/incident light
- 3 Field diaphragm
- 4 Light intensity



8.13 Aperture and Field Diaphragm

Leica DMI3000 B:

Transmitted light:

- The manual aperture diaphragm is adjusted on the condenser.
- The manual field diaphragm is adjusted on the illumination arm.

Incident light:

- The field diaphragm is set via the rotary knob (5a.11, S. 24). Diaphragm apertures of different sizes (round or angled) can be selected. (See the labeling on the rotary knob.) Round apertures are suitable for observations through the eyepieces; angled apertures are suitable for observations with CCD cameras.

Leica DMI4000 B and Leica DMI6000 B:

Both diaphragms have been set to suitable values for the current objective and contrast method at the factory.

The aperture diaphragm is controlled manually when using the manual condenser.

The field diaphragm is controlled manually when using the manual illumination arm.

- The motorized diaphragms can be adjusted at any time with the **AP** (aperture diaphragm) (118.1) and **FD** (field diaphragm) (118.3) function buttons. The values displayed on the LeicaScreen change accordingly.

The function buttons are assigned to the currently active transmitted light (TL) or incident light (IL) axis.

Microscopes with structured illumination are not equipped with a field diaphragm.



Caution: The old values will be overwritten by the current ones!



Caution:

When using **PH** or **DF**, the aperture diaphragm is fully open and can not be closed.

► **Adjust diaphragms via**

- fixed function buttons on stand
- variable function buttons on stand and SmartMove
- Leica Application Suite (LAS/LAS AF) Software

9. Trouble Shooting

Problem	Cause/Remedy
Stand	
The microscope does not respond.	▶ Ensure that the AC outlet has power. ▶ Ensure that the electronics box is connected to an AC outlet. ▶ Check the cable connections. ▶ Inform Service and have the supply unit fuse checked.
Illumination	
The image is completely dark.	▶ Open the shutter (→ p. 68). ▶ Check the connections of the lamp housings on the microscope (transmitted light/fluorescence) ▶ Ensure that the lamps are connected to the power supply and are not defective. ▶ Inform Service and have the ebq 100 supply unit fuse checked.
The image is unevenly or not uniformly illuminated.	▶ Remove all unneeded filters from the light path. ▶ Center the lamp (→ p. 78ff) ▶ Replace the old lamp (→ p. 45, 49ff).
The illumination flickers.	▶ Be sure that there is no loose connection at the power supply. ▶ Replace the old lamp (→ p. 45, 49ff).
The lamp does not illuminate immediately upon being switched on.	▶ The ebq 100 must be switched-on repeatedly. ▶ Hot Hg lamps should cool down before switching on again.

9. Trouble Shooting

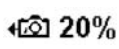
Problem	Cause/Remedy
Bright field	
The specimen can not be brought into focus.	▶ Use the correct immersion medium. ▶ Place the specimen on the stage with the coverslip facing down. ▶ Make sure that the cover glass thickness is correct and that it suits the indication on the objective. ▶ Ensure that you are using an objective with coverslip correction. ▶ Adjust the correction ring on the objective if present.
Dark Field	
No definite DF contrast is possible.	▶ Be sure that a DF objective is being used. ▶ The objective aperture is too high: maximum 0.7 for condenser S1 maximum 0.4 for condenser S23/28 If necessary, reduce the objective aperture using the iris diaphragm on the objective. ▶ Check the condenser centering.
The image is unevenly or not uniformly illuminated.	▶ The magnification is too weak. Use a higher magnification. ▶ Remove the condenser head or condenser lenses.
Undesirable stray light.	▶ Clean the specimen and neighboring lenses (→ p. 109f).

Problem	Cause/Remedy
Phase contrast	
No phase contrast is possible.	‣ The specimen is too thick. ‣ The cover glass is not placed evenly. ‣ Check the centering of the light rings (→ p. 74).
Polarization	
No polarization contrast is possible.	‣ Bring the polarizer and analyzer into cross position until they reach maximum darkness (without specimen). (→ p. 87).
Transmitted light interference contrast	
No transmitted light interference contrast is possible	‣ The specimen is too thick or too thin. ‣ Embedding medium or specimen are of birefringent material. Rotate the specimen. ‣ The difference in the refractive indices of the specimen and the embedding medium is too small. ‣ The cover glass is too thick. ‣ Check the Koehler illumination (→ p. 70). ‣ Bring the polarizer and analyzer into cross position until they reach maximum darkness (without specimen). (→ p. 87). ‣ Check whether the suitable condenser prism and corresponding objective prism are selected (manual alternative → p. 87). ‣ Ensure that the IC prisms are correctly seated (→ p. 40).

9. Trouble Shooting

Problem	Cause/Remedy
Fluorescence	
The image is completely dark (no fluorescence).	▶ Open the shutter (→ p. 68). ▶ Select the incident light axis (IL) (→ p. 66). ▶ Check your specimen, e.g. its antibody binding. ▶ Insert a new lamp (→ p. 45, p.49ff).
The fluorescence is too weak.	▶ Insert the booster (→ p. 92). ▶ Center the lamp (→ p. 78ff) ▶ Insert a new lamp (→ p. 45, p.49ff).
LeicaScreen	
Init Error!	▶ Check the cable connections. ▶ Check whether the cover of the filter disk has clicked into position. ▶ Check the installed objectives, filter cubes, etc. ▶ Switch the microscope off and back on.

13. Abbreviations and Pictograms

	Contrasting method
	Magnification
	Illumination
	Ports/Eyepiece
	Focus
	Lower focus stop not set
	Lower focus stop set
	Focus position not set
	Focus position set
	Shutter open
	Shutter closed
	Transmitted light filter
	Field diaphragm, rectangular
	Field diaphragm, round
	Aperture diaphragm
	Light distribution

13. Abbreviations and Pictograms

AP	Aperture diaphragm
BF	Bright field
COMBI	Combination method
CUBE	Fluo cube
DF	Dark field incident/transmitted light
DIC	Differential Interference Contrast
FD	Field diaphragm
FLUO	Fluorescence axis (incident light)
ICR	Interference contrast, incident light
ICT	Interference contrast, transmitted light
IL	Incident light
INT	Intensity
IMC	Integrated modulation contrast
IPH	Integrated phase contrast
PH	Phase contrast
POL	Polarization, incident/transmitted light
TL	Transmitted light

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